LeetCode 2913 — Sum of Squares of Distinct Counts

Language: Java

Question

Given a 0-indexed integer array **nums**, return the sum of the squares of the distinct counts of all subarrays of **nums**. The distinct count of a subarray is the number of distinct elements in it. A subarray is a contiguous, non-empty sequence of elements within an array.

Example

Input: nums = [1, 2, 1]
Output: 15
Explanation:
[1] → $1^2 = 1$; [2] → $1^2 = 1$; [1] → $1^2 = 1$
[1, 2] → $2^2 = 4$; [2, 1] → $2^2 = 4$; [1, 2, 1] → $2^2 = 4$
Total = 1 + 1 + 1 + 4 + 4 + 4 = 15.

Java Solution

```
import java.util.*;

class Solution {
    public int sumCounts(List<Integer> nums) {
        int n = nums.size();
        int sum = 0;

        for (int i = 0; i < n; i++) {
            Set<Integer> seen = new HashSet<>();

            for (int j = i; j < n; j++) {
                seen.add(nums.get(j));

                int distinct = seen.size();
                sum += distinct * distinct;
            }
        }

        return sum;
    }
}
```

Approach

Start each subarray at index i and extend its end index j. A HashSet tracks the distinct values in the current subarray. After adding nums[j], square the set size and add it to sum.

Complexity

Time: $O(n^2)$ subarrays, with average O(1) HashSet operations per element.
Auxiliary space: O(n) for the set.